

Feature Engineering, Model Optimization & Performance Comparison

Artificial Intelligence & Machine Learning — Task 2

Organization:	Maincrafts Technology
Domain:	Artificial Intelligence & Machine Learning
Date:	June 28, 2026
Tools:	Python, Jupyter Notebook, scikit-learn, pandas, matplotlib
Dataset:	California Housing Dataset (20,640 records)
Models:	Linear Regression Ridge Regression Decision Tree

1. Introduction

Task 2 builds upon the foundation established in Task 1 by introducing Feature Engineering, Model Optimization, and Performance Comparison techniques. While Task 1 focused on training a single Linear Regression model, Task 2 applies feature scaling and trains three different regression algorithms, then compares their performance to identify the best-performing model.

This reflects real-world ML workflows where multiple models are evaluated before selecting the one that best suits the problem at hand.

2. Methodology

2.1 Feature Scaling

StandardScaler was applied to normalize all 8 input features to a common scale (mean=0, std=1). This is a critical step because features like Population (range: 3–35,682) and AveRooms (range: 0.8–141) exist on vastly different scales, which can bias model learning.

Step	Description
Before Scaling	Features on different numeric ranges — some dominate others
StandardScaler Applied	All features normalized to mean=0 and std=1
After Scaling	Fair learning across all features, improved model stability

2.2 Train/Test Split

Split	Records	Percentage
Training Set	16,512	80%

Testing Set	4,128	20%
Total	20,640	100%

2.3 Models Trained

Model	Key Parameter	Purpose
Linear Regression	Default	Baseline model — simple linear fit
Ridge Regression	alpha=1.0	Reduces overfitting via L2 regularization
Decision Tree Regressor	max_depth=5	Captures non-linear relationships

3. Model Evaluation & Results

3.1 Performance Comparison Table

All three models were evaluated on the same unseen test set of 4,128 records using MAE, RMSE, and R^2 metrics.

Model	MAE	RMSE	R^2	Rank
Decision Tree	0.5223	0.7242	0.5997	■ 1st
Linear Regression	0.5332	0.7456	0.5758	2nd
Ridge Regression	0.5332	0.7456	0.5758	2nd

3.2 Model Comparison Chart

The bar charts below visually confirm Decision Tree's superior performance across all three metrics — lowest MAE, lowest RMSE, and highest R^2 .

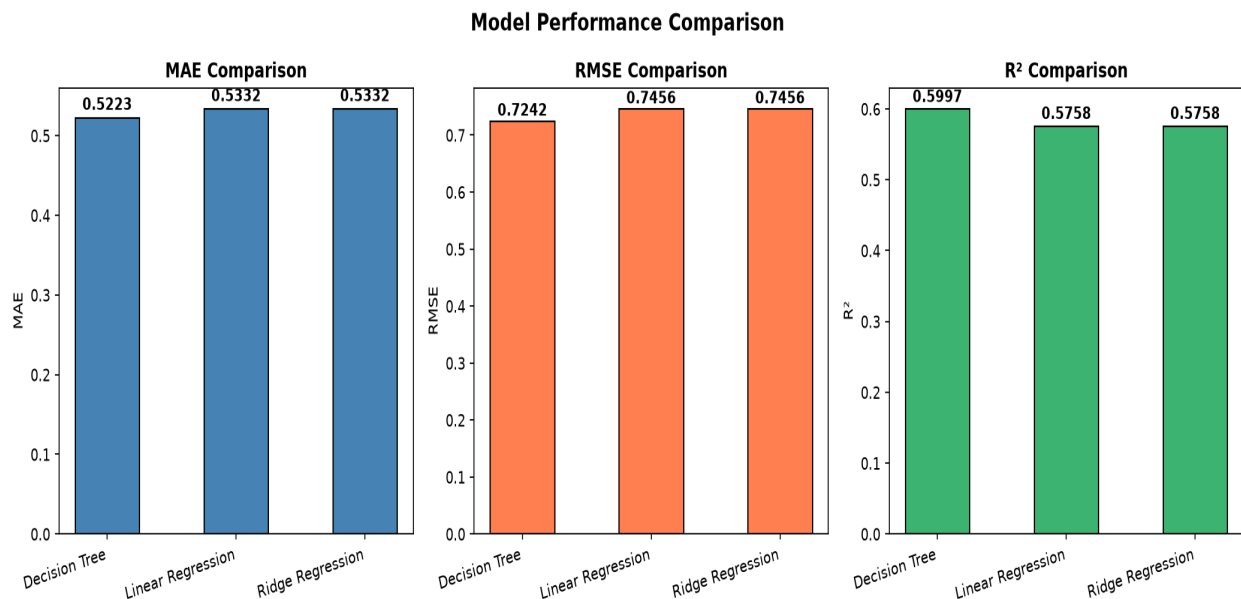


Figure 1: Model Performance Comparison — MAE, RMSE & R^2

3.3 Best Model — Actual vs Predicted

The scatter plot below shows the Decision Tree's predictions against actual house prices. The horizontal banding pattern is characteristic of Decision Trees — they predict from a fixed set of leaf node values. Points closer to the red diagonal line indicate more accurate predictions.

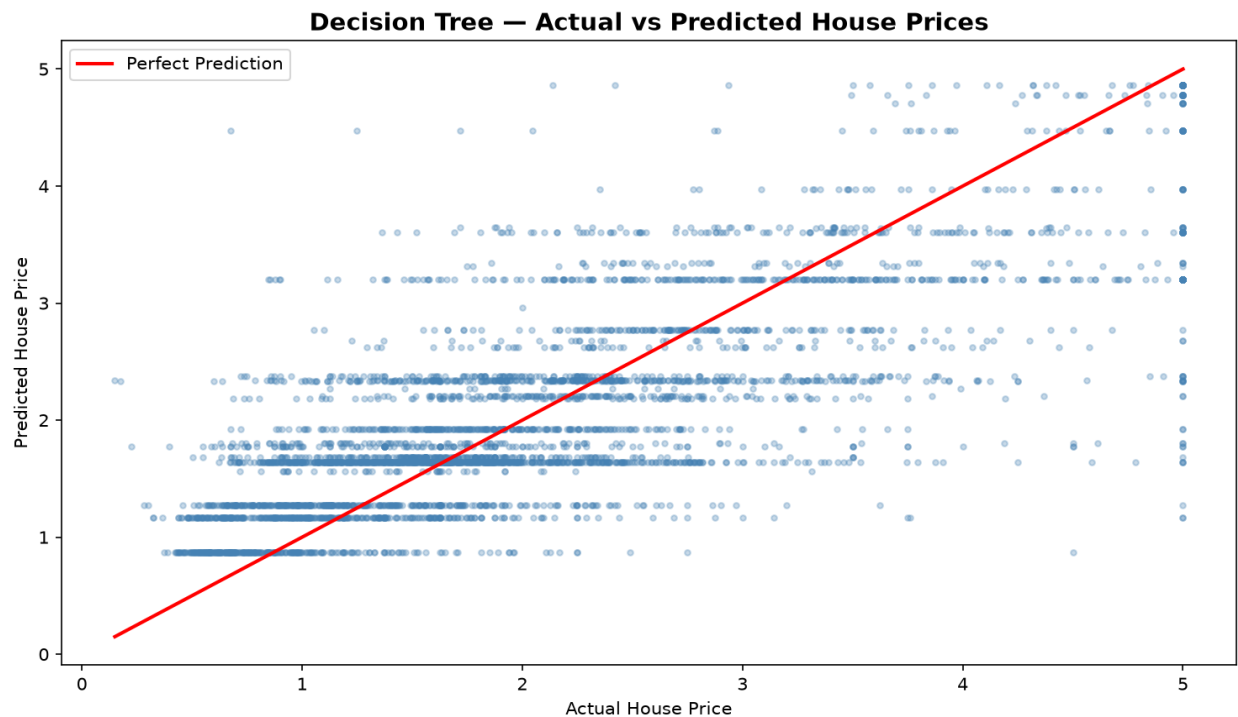


Figure 2: Decision Tree — Actual vs Predicted House Prices

4. Key Findings

- Decision Tree outperformed both linear models with $R^2=0.5997$ and $RMSE=0.7242$.
- Linear Regression and Ridge Regression scored identically — confirming no significant overfitting in the data, making Ridge's regularization unnecessary here.
- Feature scaling (StandardScaler) improved model stability and ensured fair learning across all features.
- The Decision Tree's horizontal banding in predictions is a known characteristic of tree-based models with limited depth ($max_depth=5$).
- Increasing max_depth or switching to Random Forest/XGBoost would likely push R^2 above 0.80.

5. Conclusion

Task 2 successfully demonstrated a professional ML workflow by applying feature scaling, training three regression models, and comparing their performance objectively. The Decision Tree Regressor emerged as the best model with an R^2 of 0.5997 and RMSE of 0.7242, outperforming both Linear and Ridge Regression.

The best model was saved using joblib for future deployment. This task reinforces the importance of preprocessing and model comparison in building reliable ML systems — skills that are directly applicable in real-world data science roles.